

Node attributes and performance measures

Supply Chain Node Attributes		
Sr	Attribute/Feature	Description
1	Name	Name of the supply chain (SC) node
2	ID	Unique identifier for the node (alphanumeric)
3	Type	Type/category of the SC node (e.g., manufacturer, distributor)
4	Location	Geographic location (latitude, longitude) of the node
5	Inventory	Whether the node has inventory storage capability
6	Inventory Type	Type of inventory (e.g., finished goods, raw materials)
7	Inventory Replenishment Policy	Replenishment policy (e.g., periodic, EOQ)
8	Inventory Capacity	Maximum inventory level that can be stored
9	Inventory Reorder Level	Threshold to trigger replenishment
10	Inventory Order-up-to Level	Target level to which inventory is raised when a replenishment order is placed
11	Inventory Safety Stock	Buffer stock held to absorb demand and lead time variability
12	Inventory Holding Cost(s)	Cost incurred per unit for holding inventory
13	Storage Cost	Cost of operating the storage facility, such as refrigerated storage in cold chains
14	Inventory Reorder Period	Time interval for placing replenishment orders
15	Inventory Product(s)	List of products that are stored in the inventory
16	Order Cost	Cost incurred per order placed to the supplier to replenish the inventory
17	Production Cost	Cost to produce a unit/batch of product (node-specific attribute)
18	Establishment Cost	Cost of setting up the node (building the production plant or warehouse)
19	Depreciation Cost	Decrease in product value over time due to wear, tear, or unfavorable market conditions
20	Backorder Penalty	Penalty cost for delayed orders
21	Probability of Node Failure	Probability of this node being unavailable due to power outage, equipment breakdown, or labour disruption
22	Disruption Duration	Time for which the node remains unavailable after a disruption
23	Recovery Duration	Time taken by the node to resume normal operation after a disruption
24	Disruption Cost	Cost incurred per disruption at this node, including the cost of recovery
Supply Chain Node Performance Measures		
Sr	Performance Measure	Description
1	Throughput	Number of product units sold per unit of time (day/month/quarter)
2	Total sales/net sales	Total number of items sold over a given time period
3	Net profit	Net profit generated over a given time period (current month/quarter)
4	Average net profit	Average net profit per time unit (day/month/quarter)
5	Inventory levels	Inventory levels at regular time intervals (daily/monthly)

Node attributes and performance measures

6	Total node cost	Total node cost (includes manufacturing, storing, moving, loading, unloading, and transporting)
7	Average node cost	Average node cost per unit time
8	Demand Placed	Number of orders (and product quantity) placed by this node
9	Demand Received	Number of orders (and product quantity) received at this node
10	Fulfillment Received	Number of orders (and product quantity) fulfilled for this node
11	Demand Fulfilled (Sales)	Number of orders fulfilled by this node
12	Customer Service Level	Percentage of demand fulfilled
13	Shortage	Number of orders not fulfilled by this node due to unavailability
14	Inventory Waste	Inventory waste due to product expiration (applicable for perishable products)
15	Disruption Waste	Quantity and monetary value of inventory destroyed by a disruption at this node
16	Inventory Carry Cost	Cost of storing each product item at the node
17	Inventory Spend Cost	Expenses for replenishing the inventory
18	Transportation Cost	Transportation cost paid for replenishment orders by this node
19	Revenue	Total income generated by this node
20	Production Level	Quantity of goods produced in a given time period
21	Supply Reliability	Probability of reliable supply at the node
22	Node Resilience	Node's ability to recover from disruption
23	Node Criticality	Refers to the importance of the node (e.g., a node that could cut the main flow of product if it is down)
24	Inventory Shortfall	Shortage amount (unmet demand quantity)